

MODULE 10

Seaborn

Topics Covered

- Lecture 1 — Seaborn Overview
- Lecture 2 — Relational Plot Introduction
- Lecture 3 — Lineplot, Scatter & FacetGrid
- Lecture 4 — Distribution Plots: histplot, kdeplot, rugplot
- Lecture 5 — Bivariate Histogram, KDE & Matrix Plots
- Lecture 6 — Categorical Plots
- Lecture 7 — Regression Plots & Multi Plots

Lecture 1: Seaborn Overview

Seaborn is a statistical visualisation library built on top of Matplotlib. It produces beautiful, informative plots with far less code, and integrates directly with Pandas DataFrames.

❑ Seaborn vs Matplotlib

Matplotlib: full control, verbose — you build everything manually.
Seaborn: higher-level, opinionated defaults — beautiful plots in 1–2 lines.
They work together: use Seaborn to plot, Matplotlib to customise (titles, size, labels).

Installation & Import

```
pip install seaborn

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

# Set global theme
sns.set_theme(style='darkgrid') # applies to all subsequent plots
```

Built-in Datasets

Seaborn ships with several classic datasets — perfect for practice:

```
tips = sns.load_dataset('tips') # restaurant tips
titanic = sns.load_dataset('titanic') # Titanic passengers
iris = sns.load_dataset('iris') # flower measurements
planets = sns.load_dataset('planets') # exoplanet data
flights = sns.load_dataset('flights') # monthly air passengers
penguins= sns.load_dataset('penguins') # penguin measurements

print(tips.head())
print(tips.shape) # (244, 7)
```

Plot Categories in Seaborn

Category	Plot Types Use Case
Relational	lineplot, scatterplot — relationships between numeric variables
Distribution	histplot, kdeplot, rugplot, ecdfplot — how data is spread
Categorical	barplot, boxplot, violinplot, stripplot, swarmplot — numeric vs category
Matrix	heatmap, clustermap — grid of values (correlation, pivot)
Regression	regplot, lmpplot — scatter + linear fit
Multi-plot	FacetGrid, PairGrid, pairplot, jointplot — combine many plots

Themes & Palettes

Setting	Options
Style (sns.set_style)	'darkgrid' 'whitegrid' 'dark' 'white' 'ticks'
Context (sns.set_context)	'paper' 'notebook' 'talk' 'poster' (controls font/line scale)
Palette (palette=)	'deep' 'muted' 'pastel' 'bright' 'dark' 'colorblind' any matplotlib cmap

```

sns.set_theme(style='whitegrid', context='notebook', palette='deep')

# Preview a palette
sns.color_palette('viridis', 6)    # 6 colours from viridis

```

Lecture 2: Relational Plot Introduction

Relational plots show relationships between two or more numeric variables. The main function is `sns.relplot()` — a figure-level function that can create both scatter and line plots with faceting support.

scatterplot

```

tips = sns.load_dataset('tips')

# Basic scatter
sns.scatterplot(data=tips, x='total_bill', y='tip')
plt.title('Total Bill vs Tip')
plt.show()

# Encode more dimensions with hue, size, style
sns.scatterplot(data=tips, x='total_bill', y='tip',
                hue='sex',          # colour by category
                size='size',        # point size by value
                style='smoker',     # marker shape by category
                alpha=0.8)

plt.show()

```

relplot — Figure-Level Scatter

```

# relplot wraps scatterplot with faceting
sns.relplot(data=tips, x='total_bill', y='tip',
            hue='sex', col='time',    # separate plots by 'time'
            kind='scatter', height=4, aspect=0.8)

plt.show()

```

Common Aesthetic Parameters

Parameter	Effect
hue	Map a variable to colour — works with both categorical and numeric
size	Map a variable to marker/line size
style	Map a variable to marker shape or line style
col	Split into multiple columns of subplots by a variable

Parameter	Effect
row	Split into multiple rows of subplots by a variable
palette	Colour palette for hue mapping
alpha	Transparency of points
markers	List or dict of marker shapes for style mapping

Lecture 3: Lineplot, Scatter & FacetGrid

lineplot

Seaborn's lineplot automatically aggregates repeated x-values and shows a confidence interval (shaded band) by default.

```
flights = sns.load_dataset('flights')

# Single line
sns.lineplot(data=flights, x='year', y='passengers')
plt.title('Annual Passengers'); plt.show()

# Multiple lines with hue
sns.lineplot(data=flights, x='year', y='passengers',
             hue='month',      # one line per month
             palette='tab10')
plt.show()

# Disable confidence interval
sns.lineplot(data=flights, x='year', y='passengers',
             errorbar=None)
```

FacetGrid — Custom Multi-Plot Grids

FacetGrid is the most flexible way to create a grid of plots based on values in one or more categorical variables.

```
tips = sns.load_dataset('tips')

# Create a grid: rows=smoker, cols=time
g = sns.FacetGrid(tips, col='time', row='smoker', height=4)

# Map a plot type onto each cell
g.map(sns.scatterplot, 'total_bill', 'tip', alpha=0.6)
g.add_legend()
plt.show()

# FacetGrid with histogram
g = sns.FacetGrid(tips, col='day', col_wrap=2, height=3)
g.map(sns.histplot, 'total_bill', bins=15)
plt.show()
```

□ Axes-level vs Figure-level functions

Axes-level (sns.scatterplot, sns.histplot): draw on a single Axes, compatible with plt.subplots().

Figure-level (sns.relplot, sns.displot, sns.catplot): manage their own figure + FacetGrid, support col/row splitting.

Use figure-level for quick faceting, axes-level for embedding in complex layouts.

Lecture 4: Distribution Plots — histplot, kdeplot, rugplot

Distribution plots show how a variable's values are spread. Seaborn's displot() is the figure-level wrapper; histplot, kdeplot, and rugplot are the axes-level functions.

histplot

```
penguins = sns.load_dataset('penguins')

# Basic histogram
sns.histplot(data=penguins, x='flipper_length_mm', bins=20)
plt.show()

# With KDE overlay
sns.histplot(data=penguins, x='flipper_length_mm',
             kde=True, bins=25, color='steelblue')
plt.show()

# Stacked by category
sns.histplot(data=penguins, x='flipper_length_mm',
             hue='species', multiple='stack', bins=25)
plt.show()
```

multiple= parameter	Effect on overlapping groups
'layer'	Default — overlaid transparently
'dodge'	Side-by-side bars per bin
'stack'	Stacked bars
'fill'	Stacked, normalised to 1 (shows proportions)

kdeplot — Kernel Density Estimate

Smooths the distribution into a continuous curve. Useful for comparing shapes of distributions.

```
# Basic KDE
sns.kdeplot(data=penguins, x='flipper_length_mm')
plt.show()

# Multiple groups
sns.kdeplot(data=penguins, x='flipper_length_mm',
            hue='species', fill=True, alpha=0.4, common_norm=False)
plt.show()

# Adjust bandwidth (smoothing)
sns.kdeplot(data=penguins, x='body_mass_g', bw_adjust=0.5) # sharper
sns.kdeplot(data=penguins, x='body_mass_g', bw_adjust=2.0) # smoother
```

rugplot

Draws a tick mark at each data point along the axis — shows the exact location of every observation. Usually combined with histplot or kdeplot.

```
# Standalone rugplot
sns.rugplot(data=penguins, x='flipper_length_mm')
plt.show()

# Combined with KDE — shows both density and individual points
sns.kdeplot(data=penguins, x='flipper_length_mm',
            hue='species', fill=True, alpha=0.3)
sns.rugplot(data=penguins, x='flipper_length_mm',
            hue='species', height=0.06)
plt.show()
```

displot — Figure-Level Distribution

```
# Histogram with faceting
sns.displot(data=penguins, x='flipper_length_mm',
            col='species', kind='hist', kde=True, height=4)
plt.show()

# KDE faceted
sns.displot(data=penguins, x='body_mass_g',
            hue='sex', col='species', kind='kde', fill=True)
plt.show()
```

Lecture 5: Bivariate Histogram, KDE & Matrix Plots

Bivariate histplot

A 2D histogram shows the joint distribution of two continuous variables using colour intensity.

```
tips = sns.load_dataset('tips')

# 2D histogram
sns.histplot(data=tips, x='total_bill', y='tip', bins=20, cmap='Blues')
plt.title('Joint Distribution: Bill vs Tip')
plt.show()
```

Bivariate kdeplot

```
# 2D KDE — contour lines show density
sns.kdeplot(data=tips, x='total_bill', y='tip',
            fill=True, cmap='Reds', thresh=0.05)
plt.show()

# KDE over scatter
sns.scatterplot(data=tips, x='total_bill', y='tip', alpha=0.3, color='grey')
sns.kdeplot(data=tips, x='total_bill', y='tip', color='navy', linewidths=1.5)
plt.show()
```

jointplot — Scatter + Marginals

Combines a central bivariate plot with marginal distributions on the axes.

```
# Default: scatter in centre, histograms on margins
sns.jointplot(data=tips, x='total_bill', y='tip')
plt.show()

# KDE version
sns.jointplot(data=tips, x='total_bill', y='tip',
              kind='kde', fill=True)
plt.show()

# hex bin
sns.jointplot(data=tips, x='total_bill', y='tip', kind='hex')
plt.show()

# With hue
sns.jointplot(data=tips, x='total_bill', y='tip',
              hue='sex', kind='scatter')
plt.show()
```

heatmap — Matrix Plot

Visualises a 2D matrix of values using colour — ideal for correlation matrices and pivot tables.

```
import seaborn as sns
import matplotlib.pyplot as plt

# Correlation heatmap
corr = tips[['total_bill', 'tip', 'size']].corr()

sns.heatmap(corr,
            annot=True,          # show values in each cell
            fmt='.2f',          # 2 decimal places
            cmap='coolwarm',    # diverging colormap
            vmin=-1, vmax=1,    # fix colour scale
            linewidths=0.5,
            square=True)
plt.title('Correlation Matrix')
plt.show()

# Pivot table heatmap
flights = sns.load_dataset('flights')
pivot = flights.pivot(index='month', columns='year', values='passengers')
sns.heatmap(pivot, annot=True, fmt='d', cmap='YlOrRd', linewidths=0.3)
plt.title('Monthly Flights by Year')
plt.show()
```

heatmap parameter	Description
annot	True — write data values in each cell
fmt	Format string for annotations: 'd' int, '.2f' float, '.0%' percent
cmap	Colormap: 'coolwarm' for diverging, 'Blues'/'YlOrRd' for sequential
vmin / vmax	Pin colour scale endpoints
linewidths	Width of lines between cells
mask	Boolean array — mask cells you don't want shown

heatmap parameter	Description
square	True — force square cells

Lecture 6: Categorical Plots

Categorical plots compare a numeric variable across different categories. `catplot()` is the figure-level function; the axes-level functions include `barplot`, `boxplot`, `violinplot`, `stripplot`, and `swarmplot`.

barplot & countplot

```
tips = sns.load_dataset('tips')

# barplot — shows mean (and CI) of numeric variable per category
sns.barplot(data=tips, x='day', y='total_bill',
            hue='sex', palette='Set2')
plt.title('Average Bill by Day & Gender')
plt.show()

# countplot — counts occurrences of each category
sns.countplot(data=tips, x='day', hue='sex', palette='Set2')
plt.title('Count of Records by Day')
plt.show()
```

boxplot

Shows median, IQR (box), whiskers, and outliers. Best for comparing distributions across categories.

```
sns.boxplot(data=tips, x='day', y='total_bill',
            hue='smoker', palette='pastel')
plt.title('Bill Distribution by Day')
plt.show()

# Horizontal boxplot
sns.boxplot(data=tips, y='day', x='total_bill', orient='h')
plt.show()
```

violinplot

Combines a boxplot with a KDE — shows both the summary statistics and the full distribution shape.

```
sns.violinplot(data=tips, x='day', y='total_bill',
               hue='sex', split=True, # mirror halves by hue
               palette='muted', inner='quart')
plt.title('Bill Distribution (Violin)')
plt.show()
```

stripplot & swarmplot

```
# stripplot — scatter all individual points, jittered
sns.stripplot(data=tips, x='day', y='tip', jitter=True,
              hue='sex', dodge=True, alpha=0.6)
plt.show()
```



```
# swarmplot — non-overlapping points (slower on large data)
sns.swarmplot(data=tips, x='day', y='tip', hue='sex', dodge=True)
plt.show()

# Layer box + strip for best of both
sns.boxplot(data=tips, x='day', y='tip', color='lightblue')
sns.stripplot(data=tips, x='day', y='tip', color='navy', alpha=0.4, size=3)
plt.show()
```

catplot — Figure-Level Categorical

```
# Faceted boxplot: col per time, row per smoker
sns.catplot(data=tips, x='day', y='total_bill',
            hue='sex', col='time', kind='box',
            height=4, aspect=0.7)
plt.show()
```

Plot	Best For
barplot	Compare means across categories (shows confidence interval)
countplot	Count how many records fall in each category
boxplot	Compare distributions — median, spread, outliers
violinplot	Like boxplot but also shows distribution shape (KDE)
stripplot	Show all individual data points (small to medium datasets)
swarmplot	Like strip but non-overlapping — best for small datasets

Lecture 7: Regression Plots & Multi Plots

regplot — Scatter + Regression Line

Draws a scatter plot with a fitted regression line and a confidence interval band. Simple and powerful for visualising linear relationships.

```
tips = sns.load_dataset('tips')

# Basic regression plot
sns.regplot(data=tips, x='total_bill', y='tip',
            scatter_kws={'alpha':0.5, 'color':'steelblue'},
            line_kws={'color':'red', 'linewidth':2})
plt.title('Bill vs Tip with Regression Line')
plt.show()

# Polynomial regression (degree 2)
sns.regplot(data=tips, x='total_bill', y='tip', order=2)
plt.show()

# Logistic regression (for binary y)
titanic = sns.load_dataset('titanic')
sns.regplot(data=titanic, x='age', y='survived', logistic=True,
            scatter_kws={'alpha':0.2})
plt.show()
```

Implot — Figure-Level Regression

Implot wraps regplot with FacetGrid support — use it when you want regression lines split by categories.

```
# Separate regression lines by hue
sns.lmplot(data=tips, x='total_bill', y='tip',
           hue='smoker', palette='Set1')
plt.show()

# Faceted by col and row
sns.lmplot(data=tips, x='total_bill', y='tip',
           col='time', row='sex', height=3)
plt.show()
```

regplot vs Implot	Key Difference
regplot	Axes-level — plots on a single Axes, use with plt.subplots()
Implot	Figure-level — manages its own FacetGrid, supports col/row/hue faceting

pairplot — All-vs-All Matrix

Plots pairwise relationships between all numeric columns in a DataFrame. Scatter on off-diagonals, distribution on diagonal.

```
iris = sns.load_dataset('iris')

# Default pairplot
sns.pairplot(iris, hue='species', palette='Set2')
plt.show()

# Customise diagonal and off-diagonal plots
sns.pairplot(iris, hue='species',
             diag_kind='kde',      # KDE on diagonal instead of hist
             plot_kws={'alpha':0.5},
             diag_kws={'fill':True})
plt.show()

# Only specific columns
sns.pairplot(iris, vars=['sepal_length', 'petal_length', 'petal_width'],
             hue='species')
plt.show()
```

PairGrid — Full Customisation

```
g = sns.PairGrid(iris, hue='species')

g.map_upper(sns.scatterplot)    # upper triangle: scatter
g.map_lower(sns.kdeplot)       # lower triangle: KDE
g.map_diag(sns.histplot)       # diagonal: histogram
g.add_legend()
plt.show()
```

Quick Reference — Seaborn Plot Chooser

Goal	Best Seaborn Function
Trend over time (one line)	<code>sns.lineplot</code>
Relationship between two numeric vars	<code>sns.scatterplot</code> or <code>sns.regplot</code>
Distribution of one variable	<code>sns.histplot(kde=True)</code> or <code>sns.kdeplot</code>
Compare distributions across groups	<code>sns.boxplot</code> or <code>sns.violinplot</code>
Count records per category	<code>sns.countplot</code>
Compare means per category	<code>sns.barplot</code>
All raw data points per category	<code>sns.stripplot</code> or <code>sns.swarmplot</code>
Correlation matrix	<code>sns.heatmap(df.corr())</code>
Joint bivariate distribution	<code>sns.jointplot</code>
All pairwise relationships	<code>sns.pairplot</code>
Faceted plots	<code>sns.FacetGrid</code> or figure-level functions

End of Module 10 Notes — Generative AI Course